



## **NRCan CATCH Database**

Canadian Thermochronology National Data Platform



GDAC-SA Advanced Analytics Platform RFQ  
Lithodat Pty Ltd  
ABN: 63 627 008 904

**Project Overview:** The Canadian ThermoCHronology (CATCH) database is a \$400,000 CAD national platform integrating 51 years of thermochronology data. Developed by Lithodat for Natural Resources Canada's Geological Survey, CATCH demonstrates direct alignment with GDAC-SA requirements through proven capability in multi-decade data integration, legacy data digitization, and FAIR-compliant national database architecture delivered on-time and on-budget.

## 1. Overview

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The CATCH database is Canada's first unified national repository for low-temperature thermochronology data, integrating 1,427 samples from 91 publications spanning 1970-2021. Developed by Lithodat Pty Ltd (Fabian Kohlmann, Malcolm McMillan) with the Geological Survey of Canada (Dr. Jeremy Powell, Dr. Dawn Kellett), CATCH demonstrates national-scale data standardization directly applicable to GDAC-SA's multi-agency integration requirements.

**Performance:** \$400,000 CAD, 18 months (October 2021 - March 2023), delivered on-time and on-budget. Platform scale: 996 fission-track datapoints, 762 (U-Th)/He datapoints, 16,277 single grain measurements, published via DOI 10.4095/pdreqf905k and accessible through AusGeochem.

## 2. History & Funding

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Natural Resources Canada's Geological Survey (GSC) is Canada's national geoscience organization. Prior to CATCH, Canada had no central thermochronology database despite decades of published research—analogous to GDAC-SA's challenge of fragmented geological survey data.

In October 2021, GSC awarded Lithodat a \$400,000 CAD contract to design Canada's first national thermochronology data model, harvest all published research (1970-2021), and integrate with AusGeochem's FAIR platform. The project required systematic harmonization of legacy laboratory outputs from multiple provinces, automated QA/QC pipeline development, and extensive legacy data digitization from analogue paper reports and non-vectorized PDFs.

Deliverables included national data schema accommodating four thermochronometer types (apatite/zircon fission-track, apatite/zircon (U-Th)/He), standardization of 51 years of provincial data, complete harvesting from 91 publications, and integration with international reporting standards.

## 3. Key Collaborators

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**Client:** Natural Resources Canada - Geological Survey of Canada (equivalent to Saudi Geological Survey).

**Project Leadership:** Dr. Jeremy Powell (GSC project lead, first author), Dr. Dawn Kellett (GSC co-author), demonstrating government partnership excellence.

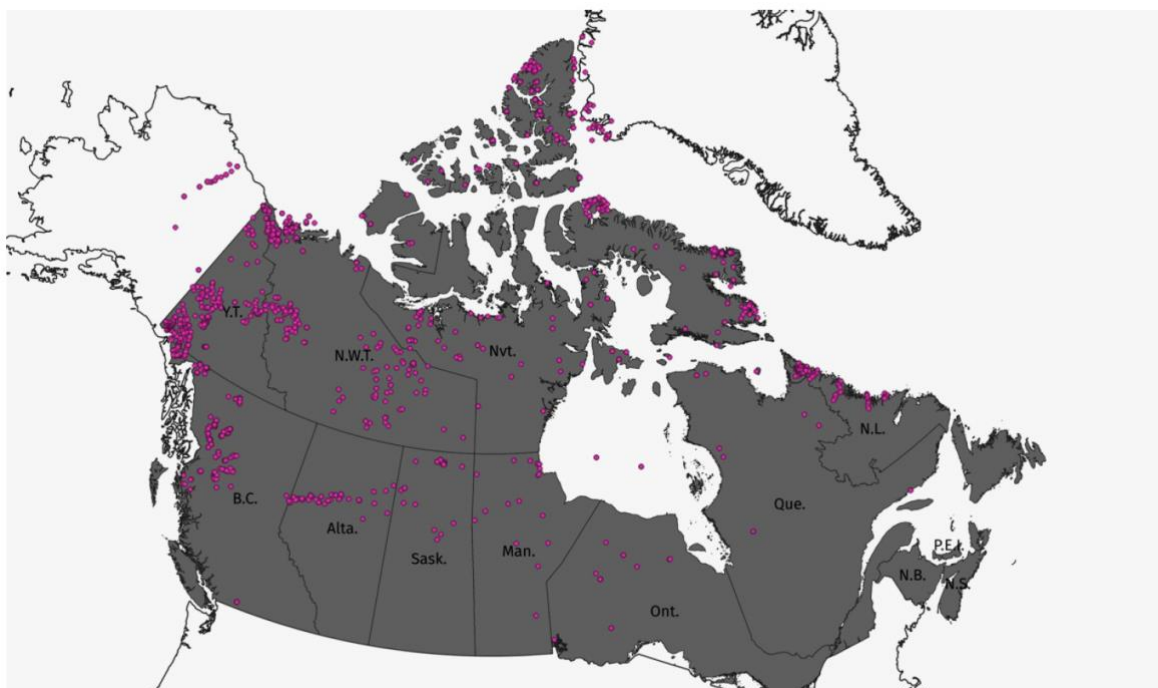
**Provincial Integration:** All Canadian provincial geological surveys contributed data requiring standardization across boundaries—demonstrating multi-agency harmonization applicable to GDAC-SA.

**Platform:** AusGeochem provides FAIR data access, enabling international research community access through open-source platform.

## 4. Platform Capabilities

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CATCH employs purpose-built data structure accommodating four thermochronometer types (Apatite/Zircon Fission Track, Apatite/Zircon (U-Th)/He) plus thermal history modeling. Canada's first unified national thermochronology schema handles complex kinetic parameters, accommodates legacy laboratory outputs, and provides ML-compatible, cloud-ready structure.



**Figure 1: CATCH database northern Canada compilation showing 1,427 sample locations with low-temperature thermochronology data (colored by geologic age: red = Precambrian, green = Paleozoic, blue = Mesozoic, yellow = Cenozoic). Demonstrates national-scale geospatial data integration with complete metadata capture for 51 years of published research. Data accessible via AusGeochem platform.**

**Advanced Features:** Single grain data preservation (16,277 measurements), fission-track length capture (27,848 measurements) from digitized histograms, automated QA/QC pipelines validated by GSC scientists, comprehensive provenance tracking, georeferencing workflow, and thermal history modeling integration.

## 5. FAIR Data & Project Performance

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**FAIR Principles:** Findable via DOI (10.4095/pdreqf905k) in GSC Open File series; Accessible through AusGeochem platform with no authentication barriers; Interoperable using community-agreed schemas; Reusable with complete metadata and ML-compatible structure.

**On-Time, On-Budget:** 18-month timeline met (October 2021 - March 2023), \$400,000 CAD managed with complete accountability, demonstrating reliable government partnership.

**National-Scale Achievement:** Standardized data across entire Canadian territory, harmonized provincial inconsistencies, integrated 51 years of data (1970-2021), harvested all published research from 91 sources—demonstrating systematic multi-decade integration applicable to GDAC-SA.

**Technical Excellence:** Automated data ingestion for multiple formats (digital, analogue paper, non-vectorized PDFs), QA/QC validation by GSC staff, and cloud-ready infrastructure supporting advanced analytics.

## 6. Publications & Recognition

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**Powell, J.W., Kellett, D.A., Kohlmann, F., and McMillan, M. (2025).** Canadian ThermoCHronology (CATCH) database: northern Canada compilation. *Geological Survey of Canada, Open File 9225*, 1 .zip file. DOI: [10.4095/pdreqf905k](https://doi.org/10.4095/pdreqf905k)

*Official government publication documenting Lithodat's design of Canada's first unified thermochronology data model and curation of 51 years of legacy data. Demonstrates multi-format extraction pipelines, automated QA/QC frameworks, and FAIR data implementation at national scale. The project's scope and methodologies are directly transferable to GDAC-SA's requirements for integrating historical geological survey data with modern analytics infrastructure.*

**For verification of project performance and technical details, contact:**

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